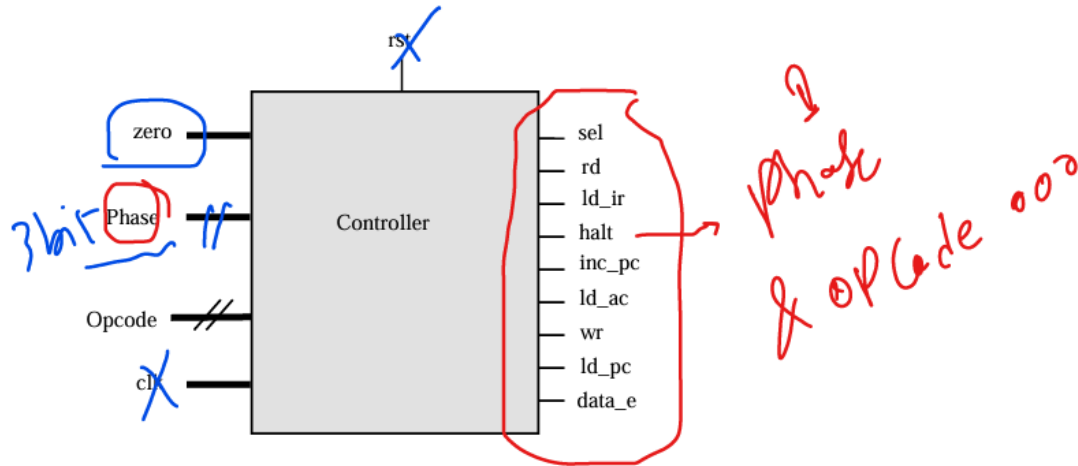


Lab 4: Modeling a Controller

Objective: To use the Verilog case statement to describe a controller.

The controller generates all control signals for the VeriRISC CPU. The operation code, fetch-and-execute phase, and whether the accumulator is zero determine the control signal levels.



Specifications

- ♦ The controller is clocked on the rising edge of ~~clk~~.
- ♦ ~~rst~~ is synchronous and active ~~high~~.
- ♦ ~~zero~~ is an **input** which is 1 when the CPU accumulator is zero and 0 otherwise.
- ♦ opcode is a 3-bit **input** for CPU operation, as shown in the following table.

Note: It was implemented but with asynchronous reset.

Waveform:

